

Electrical BUS Literature Review

Overview of the Electrical BUS Subsystem Document Revisions

The Electrical BUS subsystem's goal is to provide the CubeSat with a centralized point of control, which at a high level will involve observing indicators and driving actuators. More specifically, Electrical BUS will interact with the PAYload, RF, and EPS subsystems to manage the state of affairs on the CubeSat. Operations of the electrical subsystem may include directing the PAYload subsystem to send data to the RF subsystem for down-linking, and acquiring uplink data from RF to perform a reboot.

Version	Date	Author	Description
0.1	July 2026	Boris	Initial
0.2	July 2026	Jennifer	Expanded protocol, interconnect, and COTS trade studies.
0.3	July 2026	Boris	Protection considerations
0.4	July 2026	Sayali	Memory and peripherals
0.5	July 2026	Matthew	PAY connectors
0.6	July 2026	Asad	SRS-4 integration

Literature Review Methodology

As it pertains to the design, the main components that make up the electrical subsystem essentially boil down to three categories: control unit (that is, device(s) responsible for carrying out generic operations of the electrical subsystem, for example an MCU), protocol (that is, an interface for connecting two or more devices and allowing them to share data in one way or another, for example I2C), and peripherals (that is, device(s) responsible for a particular function within the Electrical BUS subsystem, for example a watchdog). This classification is not a recognized one, but has been found to be useful when conducting this literature review since the overwhelming majority of the designs can be deconstructed into components each of which would fit into one of these categories [subjective].

In terms of sources, two major categories are surveyed:

- Commercial OBCs discussed in NASA's Small-Sat State of the Art reports from 2022 to 2026 ([1, pp. 215–226], [2, pp. 225–237], [3, pp. 227–239], [4, pp. 241–253]), with trends also observed.
- OBCs built by academic design teams around Canada and beyond (considered designs listed in Appendix A)

Design Space

This section outlines the resulting design space for the Electrical BUS Subsystem based on the corresponding literature review. We establish a space of common control units, protocols and peripherals for our internal design process.

Control Units

The control-unit design space ranges from low-power microcontrollers to heterogeneous processor-FPGA systems. NASA's State of the Art reports show that most conventional CubeSat OBCs use microcontrollers or embedded microprocessors because command execution, telemetry collection, timekeeping, fault management, and subsystem coordination do not require high computational performance [1, 2, 3, 4]. Higher-performance FPGAs, application processors, and system-on-chip devices are generally introduced when the OBC must also support image processing, autonomy, high-rate payload data, or machine-learning workloads.

Table 1: Candidate OBC control-unit architectures

Architecture	Characteristics	NASA SOA Examples	Academic Examples
Single MCU or processor	One device performs command execution, telemetry, timekeeping, fault management, and subsystem control. Lowest complexity and generally lowest power.	EnduroSat OBC, ARM Cortex-M7 [2, p. 229]; AAC Clyde Space Kryten-M3, Cortex-M3 [2, p. 228].	STM32L4 PC/104 OBC; Athena TMS570; ORCASat TMS570 [5, 6, 7].
Multicore SoC	Multiple cores share one device and memory system. Provides additional performance or lockstep capability, but does not remove the SoC as a single point of failure.	Argotec FERMI, dual-core LEON3FT [2, p. 229]; SPiN MA61C, dual-core GR712RC [3, p. 236]; Xiphos Q7S, dual-core Cortex-A9 [3, p. 238].	Athena and ORCASat use lockstep-capable ARM safety microcontrollers [8, 9].
Dual redundant processors	Two processor devices provide cold, warm, or active redundancy. One processor may take control following a fault in the primary device.	Andes OBC, two Cortex-M7 processors with full cold redundancy [4, p. 230]; Bradford Binary OBC, dual Cortex-R5 with internal cold redundancy [2, p. 229]; Bradford Stellar OBC, dual PolarFire SoCs with internal cold redundancy [4, p. 230].	No directly equivalent two-device redundant architecture was identified in the academic survey.
Controller plus FPGA or DPU	A low-power controller performs critical spacecraft control while an FPGA or higher-performance processor handles payload data or acceleration. The secondary unit may be powered down when unused.	KP Labs Antelope, RM57 OBC plus Zynq UltraScale+ DPU [2, p. 230]; NOVI Gen 2, Vorago MCU plus Versal AI Edge SoC [4, p. 232].	NEUDOSE, primary AVR32 plus secondary Xilinx Zynq-7000 [10].
FPGA-centric	Control or processing is implemented using programmable logic and, optionally, a soft processor. Suitable for deterministic parallel interfaces and high-rate data paths, but more difficult to develop as a general-purpose OBC.	LASP Generic SmallSat FPGA Board, Kintex-7 [2, p. 230]; KP Labs Lion, Kintex UltraScale [2, p. 230].	No FPGA-only primary OBC was identified in the academic survey.

Table 1: Candidate OBC control-unit architectures (Continued)

Architecture	Characteristics	NASA SOA Examples	Academic Examples
Distributed controllers	Subsystem-local processors control the EPS, ADCS, communications, or payload and exchange data with a coordinating OBC. This improves fault containment but increases network and software complexity.	NASA identifies federated, distributed, heterogeneous, and mixed-criticality avionics as viable SmallSat architectures [2, p. 225] [4, p. 221–222].	Individual subsystem controllers used alongside the primary academic OBC.

All architectures will be considered during preliminary design; however, the current baseline is a centralized microcontroller-based OBC with independent watchdog and reset circuitry. This approach is the most common among the academic systems surveyed and provides the lowest-risk balance of power, determinism, software complexity, and flight heritage. An FPGA or secondary processor may be added later if payload-processing requirements cannot be met by the primary MCU.

Protocols and Interconnect

This section will summarize the survey of protocols across the NASA's SOAs as well as Canadian design teams' CubeSats in Table 2, as well as provide an overview of interconnect architectures in Table 3. Communication interfaces differ not only in throughput, but also in topology, wiring complexity, fault containment, and implementation requirements. Consequently, protocol selection should consider the electrical implementation and system architecture in addition to nominal data rate.

Table 2: Protocol and electrical-interface design space

Protocol	Representative Rate	Survey Examples	Primary Considerations
I ² C	100 kbit/s–3.4 Mbit/s	LORIS OBC [11]; identified as a common synchronous interface in all four NASA surveys [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	Low pin count and broad peripheral support; limited distance and bandwidth; shared-bus faults may block all devices. Bus capacitance and pull-up sizing constrain the usable rate as devices are added. If SDA remains stuck low, the I ² C specification recommends nine SCL pulses, followed by hardware reset or power cycling if the bus is not released [12].
SPI	Application-dependent; commonly 1–50+ Mbit/s	LORIS OBC [11]; NASA interface surveys [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	Deterministic and comparatively fast; requires separate chip-select signals and provides no standardized addressing or error handling. Clock mode, chip-select timing, and SCLK routing or fan-out must be verified for each peripheral, particularly at higher rates [13].

Table 2: Protocol and electrical-interface design space (Continued)

Protocol	Representative Rate	Survey Examples	Primary Considerations
UART	Device-dependent; commonly kbit/s to several Mbit/s	LORIS uses UART [11]; asynchronous serial interfaces are listed by NASA [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	Simple asynchronous point-to-point communication with low implementation complexity. UART does not inherently provide addressing, arbitration, acknowledgement, or error recovery, so these functions must be implemented by higher-level software [?].
RS-422	Approximately 90 kbit/s at 1,200 m to 10 Mbit/s at 4.5 m	Differential serial interfaces are listed by NASA [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	Differential simplex multidrop communication with one driver and up to ten receivers. It provides greater noise immunity and usable cable length than logic-level UART, but requires appropriate cabling and termination [14].
RS-485	Approximately 100 kbit/s at 1,200 m to 10 Mbit/s at 12 m	Differential serial interfaces are listed by NASA [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	Differential multipoint communication allowing multiple transceivers to share a bus. Termination is required for reliable higher-rate or longer links, and half-duplex implementations require driver-direction control [14, 15].
CAN / CAN FD	1 Mbit/s Classical CAN; commonly up to 8 Mbit/s CAN-FD data phase	STM32L4 PC/104 OBC [5]; CAN is listed as a common spacecraft fieldbus [1, p. 219] [2, p. 235] [3, p. 240] [4, pp. 224–225].	Differential multidrop operation, arbitration, acknowledgements, and error confinement make it suitable for EPS, ADCS, RF, and housekeeping traffic. Payload throughput is limited. The bus should be terminated at its two physical ends, with node stubs kept short to reduce reflections [16, 17].
USB 2.0	12 Mbit/s full-speed; 480 Mbit/s high-speed	LORIS OBC [11].	High COTS compatibility and throughput, but host/device roles, enumeration, connector retention, software complexity, VBUS sensing, and the board's USB power arrangement must be defined [18, 19].
Ethernet	100 Mbit/s–1 Gbit/s commonly	Listed as a network interface in each NASA survey [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	High throughput and broad software support; requires switching or routing and may require additional measures for deterministic real-time operation.

Table 2: Protocol and electrical-interface design space (Continued)

Protocol	Representative Rate	Survey Examples	Primary Considerations
SpaceWire / SpaceFibre	SpaceWire: 2–400 Mbit/s; SpaceFibre: up to 5 Gbit/s per lane and over 20 Gbit/s using multiple lanes [20, 21]	SpaceWire is listed in the 2022–2024 surveys [1, p. 219] [2, p. 235] [3, p. 240]; SpaceWire and SpaceFibre are included together in 2026 [4, p. 225].	Spacecraft-specific point-to-point networking with packet routing and link-level fault management. SpaceFibre additionally supports virtual channels, quality of service, and improved fault detection, isolation, and recovery. Both require more specialized implementation and test resources than conventional MCU buses [20, 21].
PPS (Pulse Per Second)	1 pulse/s	GNSS receivers commonly provide a configurable timing-pulse output; the ZED-F9P is one representative implementation [22].	Provides a precise synchronization edge but does not independently encode the absolute time. It is normally paired with UART or another data interface; electrical level, jitter, routing, and holdover behaviour must be considered.
SerDes / MGT	Gigabit-per-second class; XAUI provides 10 Gbit/s aggregate	Identified by NASA as high-rate mission-data interfaces [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 225].	Suitable for FPGA, payload, mass-memory, and radio data paths. Requires controlled-impedance routing, suitable connectors, reference clocks, and signal-integrity analysis.
GPIO / PWM / ADC / DAC	Application-dependent	LORIS provides GPIO, PWM, and ADC [11]; discrete and analog interfaces are listed by NASA [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 225].	Appropriate for enables, inhibits, deployment signals, and analog telemetry; creates dedicated wiring and provides little inherent fault detection.
JTAG / ISP / ICSP	Not a mission-data interface	Debug and programming interfaces are listed in the NASA surveys [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 225].	Required for programming and board-level debugging; access should be isolated or protected during flight.

Table 3: Candidate interconnect architectures

Architecture	Connection Pattern	Advantages	Limitations
Unified stack	OBC, EPS, RF, ADCS, and payload share a PC/104-style stack-through connector carrying power and common buses. The STM32L4 academic OBC uses this format [5]. NASA discusses PC/104 and stackable mezzanines in [1, p. 210], [2, pp. 226–227], [3, pp. 229–230], and [4, p. 222].	Compact, modular, rugged, and requires little harnessing. The self-stacking arrangement eliminates the need for a separate backplane or card cage [23].	Shared pinout constrains all boards; board order, connector bandwidth, and available signal lanes may limit expansion. Shared power or bus paths can also reduce fault isolation between boards.

Table 3: Candidate interconnect architectures (Continued)

Architecture	Connection Pattern	Advantages	Limitations
Backplane	Subsystem cards plug into a shared CompactPCI or SpaceVPX backplane [1, p. 210] [2, p. 226] [3, p. 229] [4, p. 222].	Rigid mechanical support, modular replacement, standardized slot and module interfaces, and high-speed routing. SpaceVPX additionally targets interoperable and fault-tolerant systems [24].	Higher mass, volume, connector cost, and chassis complexity, with additional mechanical, electrical, and interoperability verification.
Centralized point-to-point	Each subsystem has a dedicated connection to the OBC; all commands and data pass through the OBC. NASA identifies direct point-to-point and centralized architectures in [1, p. 207] and [3, p. 226].	Simple command authority, dedicated links, predictable data paths, and strong link-level isolation.	Connector and harness count increase with the number of subsystems. The OBC can become a bandwidth bottleneck and single point of failure, particularly when high-rate payload traffic passes through it [4, pp. 221–223].
Shared peer network	OBC, EPS, RF, ADCS, and payload controllers communicate as peers over CAN, Ethernet, or SpaceWire.	Reduces dedicated wiring, allows direct subsystem communication, and supports distributed control. Subsystem-local processing may reduce dependence on the primary OBC [4, pp. 221–222].	Requires addressing, arbitration or routing, synchronization, and network-level fault management. Shared-bus behaviour, node failures, and any redundancy-management strategy must be defined and verified [17, 20].
Payload pass-through	The OBC controls the payload, while high-rate data travels directly between the payload, mass memory, FPGA, and RF subsystem.	Avoids loading the housekeeping bus or copying all payload data through the primary MCU, while allowing a dedicated path to support higher throughput [4, pp. 222–223].	Requires separate control and data paths, flow control, buffering, and an OBC mechanism to isolate or reset the data path. The OBC must retain sufficient status and fault visibility to manage the path safely.
Hybrid dual-plane	A low-speed CAN or stack bus carries commands and housekeeping data, while dedicated Ethernet, SpaceWire, or SerDes links carry payload data. NASA describes this mixed topology in [4, pp. 222–223].	Combines a robust housekeeping network with scalable payload throughput. Separating control and high-rate traffic can reduce contention and improve fault containment between traffic classes.	Requires two interface classes, additional integration and verification, and clear ownership of commands, telemetry, timing, and high-rate data. Fault detection and recovery must be coordinated across both planes.

Redundancy is not necessarily a separate base topology and may instead be applied selectively to critical links, buses, or nodes within any of the architectures above. It can improve availability, but adds transceivers, wiring, connector pins, power consumption, software logic, and verification effort [4, 17].

For the mission, the interconnect as well as the protocol bouquet selection will want to target higher throughput interfaces for RF and PAY subsystems, and thus will generally be dictated by them. In the spirit of a generic design, selecting a higher-speed interface is favoured, yet this might collide with the interconnect architecture selection since certain interfaces require specific connections that may not, for example, be routed through PC/104 due to

signal integrity requirements (e.g. SpaceFibre). The stack-up requirements and/or constraints might also be driven by Mechanical subsystem. Consequently, interface selection should balance throughput with connector capability, harness length, routing, and signal-integrity constraints rather than relying on nominal data rate alone [20, 21].

Peripherals

This section reviews the common peripherals that are found on OBCs surveyed by NASA's SOAs, as well as on OBCs designed by Canadian academic design teams. Details about volatile and non-volatile memory can be found in Tables 4 and 5, respectively. Other types of peripherals are summarized in Table 6. General considerations related to peripherals revolve around external vs internal peripherals since modern day SoCs contain many of the necessary peripherals on the same chip.

Volatile Working Memory

Volatile working memory, which loses functionality when the electrical power is disrupted, acts as high-speed temporary storage. For a CubeSat, it enables real-time software tasks. The primary consideration during implementation is between internal and external memory: internal memory (built within the main processor chip) reduces component count and PCB complexity; external memory provides greater capacity but introduces additional power and routing requirements. Other features such as Error-Correcting Code (ECC) can be chosen as needed; ECC RAM, which automatically detects and corrects single-bit errors, was used by ORCASat [9].

Table 4: Volatile Memory Types, Metrics, Considerations, and Examples

Memory Type	Description, Considerations, & Examples	Estimated Speed	Estimated Capacity
SRAM (Static RAM)	Typically internal cache or memory within an SoC, used for low-latency code execution. Surveyed by NASA [1, p. 218] [2, p. 234] [3, p. 239] [4, p. 223].	10-25 ns access time [4]	4-32 Mb [4]
DRAM (Dynamic RAM)	High-density external volatile memory that uses single-transistor, single-capacitor cells. Surveyed across NASA missions [1, p. 218] [2, p. 234] [3, p. 239] [4, p. 223].	25 ns access time [4]	N/A
SDRAM (Synchronous DRAM)	High-density external memory synchronized with the system clock; utilized externally by RADSAT-SK for general runtime tasks [25].	N/A	N/A
DDR3 (Double Data Rate 3)	High-speed third-generation SDRAM; Dalhousie University's LORIS uses 256 MB DDR3 [11].	13-15 ns [26, pp. 67-70]	4 GB [26, p.1]

Non-Volatile Program and Mass Memory

Table 5: Non-Volatile Memory Types, Metrics, Considerations, and Examples

Memory Type	Description, Considerations, & Examples	Estimated Speed	Estimated Capacity
MRAM (Magnetoresistive)	High-endurance, radiation-tolerant non-volatile memory that relies on magnetic states to store data; typically used as internal or external storage for persistent system state. Used by the C3S OBC [2, p. 229] and in redundant configurations on the Andes OBC [3, p. 232] [4, p. 230].	300 ns access time [4, p. 224]	up to 8 Gb [27]

Table 5: Non-Volatile Memory Types, Metrics, Considerations, and Examples (Continued)

Memory Type	Description, Considerations, & Examples	Estimated Speed	Estimated Capacity
FRAM (Ferroelectric RAM)	Low-power, fast-write non-volatile memory (internal or external) that stores memory due to its ferroelectric properties; electric dipoles stay in place even without power [2, p. 229].	300 ns access time [4, p. 224]	up to 16 Mb [27]
NAND Flash	External storage where memory cells are connected in series; deployed in redundant configurations on the Andes OBC [3, p. 232] [4, p. 230].	25–50 μ s read access time [28]	128 Mb–2Tb [29]
NOR Flash	External storage where memory cells are connected in parallel; used by Athena [6] and ORCASat [7].	10–100 ns read access time [28]	128 Mb–2 Gb [29]

Non-volatile mass memory is persistent computer storage that retains stored data without requiring a continuous power supply, unlike volatile memory. It can store system boot code, flight software images, and long-term scientific data. Unlike the considerations for volatile memory, selection for non-volatile memory depends on many factors, including capacity, write endurance, retention, access speed, boot reliability, and susceptibility to radiation-induced corruption. Redundant boot images and error detection should be considered. After choosing a memory type, the packaging can be considered. For example, an eMMC (embedded MultiMediaCard) is a high-density external mass storage featuring an integrated flash controller; used by the C3S OBC [2, p. 229] and by LORIS (4 GB eMMC) [11].

Table 6: Common OBC peripheral classes

Peripheral Class	Examples	Primary Considerations
Watchdog, reset, and supervisor circuitry	The C3S OBC uses a multi-level watchdog [2, p. 229] [3, p. 232] [4, p. 230]; the Aerospacelab OBC includes a radiation-hardened supervisor MCU and watchdog [3, p. 232] [4, p. 229]. NASA discusses watchdogs and power cycling as fault-mitigation mechanisms [1, p. 220] [2, p. 236] [3, p. 241] [4, pp. 220–222].	An external watchdog is preferable for recovery from processor lock-up. Consider independent clocks, reset coverage, timeout configuration, communications watchdogs, and selective power cycling.
Timers and time-keeping	NASA identifies PLLs, capture/compare units, and time-processing units as common OBC peripherals [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224]. Timers are integrated within the STM32L4, TMS570, and i.MX6DL devices used by the surveyed academic OBCs [5, 8, 9, 30].	Used for scheduling, timestamping, watchdog servicing, and waveform generation. Clock drift, oscillator redundancy, synchronization, and operation during processor resets must be considered.
Analog and discrete I/O	NASA lists GPIO, ADC, and DAC peripherals in each avionics survey [1, p. 219] [2, p. 235] [3, p. 240] [4, pp. 224–225]. LORIS exposes GPIO, PWM, and ADC interfaces [11, 30].	Used for analog telemetry, enables, inhibits, deployment signals, and simple sensors. Consider voltage compatibility, input protection, ADC accuracy, safe reset states, and whether signals require isolation or redundancy.
Debug and programming interfaces	NASA lists JTAG, ISP, ICSP, BDM, BITP, and DB9 interfaces [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 225].	Required for programming, boundary scan, manufacturing tests, and debugging. Flight hardware should provide controlled physical access and prevent unintended entry into debug or programming modes.

Table 6: Common OBC peripheral classes (Continued)

Peripheral Class	Examples	Primary Considerations
Optional integrated navigation peripherals	The EnduroSat OBC is available with integrated GNSS [2, p. 229] [4, p. 230].	Integration can reduce board count and harnessing, but may couple the OBC lifecycle to a specific receiver and complicate antenna placement, RF isolation, replacement, and independent fault recovery.

[TODO: survey each class of peripherals in more detail]

Connections between PAY and BUS

Below in Table 7 is a survey of the OBC integration done by KP Labs.

Table 7: Physical connectors (surveyed from KP Labs Intuition-1 integration [31])

Connector Type	Supported Protocols	Notes
Nano-D Connectors	LVDS	Used for direct high-speed data transfer (e.g., connected to X/S-band radio).
M80 Connectors	Controller Area Network (CAN) bus	Used to connect DPU supervisor module to platform OBC.
Samtec LSHM	LVDS	Used for high-speed data streams (up to 8 Gbps) between optical instruments and data processing unit (DPU).

[TODO: review more examples]

SRS-4 Integration

The Satlab SRS-4 is a flagship software-defined S-band transceiver designed for CubeSat high-speed communications [32]. The SRS-4 serves as a primary communications radio to support high-throughput payload downlink requirements (such as large hyperspectral image files) alongside housekeeping telemetry and telecommand handling.

The transceiver functions as the satellite's primary voice-and-data translator between the On-Board Computer (OBC) and the RF antenna array. Satlab manufactures the SRS-4 board using the PC/104 form factor ($93.0 \times 87.2 \times 17.5$ mm, mass 253 g), allowing direct mechanical integration into standard CubeSat stackups via PC/104 header connectors and standoffs alongside EPS and OBC modules.

Unlike legacy fixed-function analog radios, the SRS-4 utilizes a Software-Defined Radio (SDR) architecture with on-orbit reconfigurable modulation and full-duplex operation, enabling simultaneous transmission and reception (requiring strict RF isolation between TX and RX paths).

Electrical and Logical Interfacing

The logical and protocol fabric of the SRS-4 natively integrates with standard Command & Data Handling (C&DH) firmware architectures:

- **CubeSat Space Protocol (CSP):** C&DH software running real-time operating systems (e.g., FreeRTOS) for task management can use CSP as a network-layer protocol. The SRS-4 speaks CSP natively. Operating like a

lightweight IP/UDP stack, CSP assigns node addresses to subsystems (OBC, EPS, ADCS, Radio) and routes packets using port-based services regardless of underlying physical media (CAN-bus, RS-422, I2C, or Ethernet). Flight software can issue commands such as `csp_send(node=radio, port=downlink, data)` agnostically across buses.

- **Interfacing Links:**

- **OBC ↔ SRS-4:** CAN-bus or RS-422 differential bus carrying CSP packets.
- **EPS ↔ SRS-4:** Wide-input power rail (5.1–28.8 V), granting flexibility to EPS regulation strategies.
- **SRS-4 ↔ Antenna:** SMA/MCX coaxial pigtail point-to-point RF link.
- **Antenna ↔ Ground:** Free-space S-band RF link (2200–2290 MHz downlink, 2025–2110 MHz uplink).

Subsystem Team Responsibilities

Integrating the SRS-4 into a satellite design spans multiple engineering domains:

- **Electrical Subsystem:** Mechanically slots the SRS-4 board into the PC/104 stack and wires the physical CAN/RS-422 bus and EPS power connections.
- **C&DH Subsystem:** Provides the RTOS and CSP network fabric where the SRS-4 operates as a standard CSP node on the system bus.
- **Firmware Team:** Develops device drivers using Satlab’s support library to queue image payload downlinks, adjust TX power, and poll operational telemetry (temperature, current draw, lock status).
- **Power (EPS) Subsystem:** Manages the input power supply rail, leveraging the SRS-4’s broad 5.1–28.8 V input range.
- **Mission Operations Team:** Utilizes CCSDS-compliant coding and standard PSK modulations to interface with commercial ground station networks and feed telemetry directly into the mission control framework.

COTS (Commercial Off the Shelf) Design Alternatives

Description: Determine what COTS solutions exist if we cannot produce a certain part of the subsystem in house. Find a COTS option for each in house design.

Will inform trades later on for COTS versus in house risks.

This section outlines alternative off-the-shelf components that would in part or in full fulfil the operating requirements of the subsystem.

The following component-level COTS alternatives preserve a custom Electrical BUS architecture while reducing development effort for selected functions. They are trade-study candidates rather than flight-qualified selections; mission-specific part assurance, derating, and radiation assessment remain required [33].

Table 8: Preliminary component-level COTS alternatives

Function	Example Candidates	Potential OBC Use	Main Trade Considerations
Non-volatile storage	Everspin MR25H40 SPI MRAM for critical state and logs [34], and Infineon S25HL512T serial NOR flash for firmware storage or recovery images [35].	Persistent configuration, deployment state, reset history, fault logs, firmware, and a recovery image.	Capacity, endurance, retention, write protection, boot recovery, power-loss behaviour, and radiation response.

Table 8: Preliminary component-level COTS alternatives (Continued)

Function	Example Candidates	Potential OBC Use	Main Trade Considerations
Independent fault recovery	TI TPS3431 watchdog, TPS3703 voltage supervisor, and TPS22919 load switch [36, 37, 38].	Detect processor lock-up, supervise the supply, and provide a hardware reset or power-cycle path.	Independence from the MCU, timeout and threshold selection, reset coverage, load current, default state, and control ownership.
Timing and clocking	Micro Crystal RV-3028-C7 RTC [39], plus an industrial crystal or TCXO selected using oscillator guidance [40].	Maintain timestamps through resets and provide stable MCU and interface timing.	Backup supply, drift, calibration, startup margin, temperature stability, synchronization, and PCB placement.
Differential bus interface	TI TCAN1042-Q1 CAN/CAN-FD transceiver [41] or Analog Devices ADM3065E RS-485 transceiver [42].	Implement the physical layer between MCU logic and a differential bus.	Common-mode range, delay, standby state, termination, fault protection, isolation needs, and radiation assessment.
Board-health monitoring	TI TMP117 temperature sensor and INA238 current, voltage, and power monitor [43, 44].	Provide local temperature and supply-rail telemetry for health monitoring.	Placement, range, shunt sizing, conversion time, addressing, power, failure response, and radiation behaviour.

System Architecture and Budgets

In this section we concern ourselves with system-level considerations such as power draw, protection mechanisms, and radiation tolerance for the purposes of overall design and integration with other teams.

Power draw

Based on the commercial onboard-computing systems surveyed in NASA's State of the Art reports [1, 2, 3, 4], OBC power consumption can be divided into three broad categories according to platform complexity and computational capability. The reports do not use a consistent average-power metric; reported values are variously identified as idle, static, typical, active-mode, full-load, or peak consumption. Consequently, Table 9 preserves the operating-state terminology used by the manufacturers.

The 2022 edition identifies representative onboard-computing products but does not tabulate their power consumption [1, Table 8-1, pp. 214–217]. Quantitative power values are therefore drawn primarily from the 2023, 2024, and 2026 editions [2, Table 8-1, pp. 228–233] [3, Table 8-1, pp. 230–236] [4, Table 8-4, pp. 229–234].

Table 9: OBC power-consumption trends and reported operating states

Platform Complexity	Reported Power Profile	Design Characteristics	Example Designs
Low (Bare-Metal / Control)	Minimum listed: 0.05 W Idle/steady: approximately 0.2–0.5 W Peak: generally below 1 W	Simple command and data handling, housekeeping, telemetry, and fault management. These systems are normally based on low-power microcontrollers and execute bare-metal software or a small real-time operating system. Their power consumption is relatively insensitive to computational load.	Pumpkin PPM family: 0.05 W, operating state not specified [2, p. 231] [3, p. 235] [4, p. 233]. AAC Clyde Space Kryten-M3: 0.4 W [2, p. 228] [3, p. 230] [4, Table 8-4]. C3S Electronics OBC: 0.46 W in test mode [2, p. 229] [3, p. 232] [4, p. 230]. SkyLabs NANOobc-2: < 0.9 W idle and < 1 W peak [2, p. 232] [3, p. 236].
Medium (Integrated OBC)	Idle: approximately 0.2–1 W Nominal/steady: approximately 1–3 W Peak: approximately 1.5–5 W	Typical integrated CubeSat OBCs combining command and data handling, mass-memory management, communication interfaces, fault protection, and basic payload management. Power may vary with processor utilization, interface activity, and memory access.	SPiN MA61C CubeSat: 1–1.2 W [2, p. 232] [3, p. 236] [4, Table 8-4]. AAC Clyde Space Sirius OBC & TCM: 1.3 W [2, p. 228] [3, p. 230] [4, Table 8-4]. EnduroSat OBC I: 0.2 W idle and 1.5 W peak [2, p. 229] [3, Table 8-1] [4, p. 230]. Resilient Computing RadPC: 1.5 W [2, p. 232] [3, p. 236] [4, p. 233]. SatRev OBC: 2.8 W typical and 5 W peak [4, p. 233].
High (Edge / AI / SoM)	Idle/static: approximately 5–15 W Typical/active: approximately 6–10 W Peak/full load: commonly 10–20 W, with larger payload computers reaching substantially higher values	Heterogeneous systems-on-module and payload-processing computers supporting image processing, software-defined radio, autonomous navigation, data compression, and machine-learning inference. Power is strongly dependent on processor utilization, FPGA configuration, accelerator activity, and payload duty cycle.	Novo Space GPU002AF: 6–11 W in active mode [2, p. 231] [3, p. 235] [4, p. 233]. KP Labs Leopard DPU: 7 W static and up to 10 W during deep-learning inference [2, p. 230] [3, p. 234] [4, p. 231]. Aitech S-A1760: ≤ 5 W idle, 8–10 W under a typical CUDA load, and < 20 W at full utilization [2, p. 228] [3, Table 8-1] [4, Table 8-4]. EnduroSat GPC: < 15 W idle and 130 W peak [4, p. 231].

Based on these results, a nominal power-design envelope of 1–10 W is adopted for the proposed OBC. This range encompasses conventional integrated CubeSat OBCs and moderate edge-processing workloads, while transient loads above 10 W are treated separately as peak operating conditions. The selected range is a project design boundary rather than the full commercial-hardware envelope, which extends from approximately 0.05 W for low-power microcontroller boards to 130 W peak for high-performance payload computers.

[TODO: peak loads and durations]

Protections

Protecting the electrical compenets of the BUS electronics from the harsh space environment is of paramount concern to the design. This section details the consideraitons with respect to the design and how to achieve them

Table 10: Protection considerations

Consideration	Description	Example solutions
Mission radiation environment	Radiation exposure depends on orbit, inclination, mission duration, shielding, and solar activity. Parts should be selected against a mission-specific radiation analysis with an appropriate design margin [1, pp. 219–220] [2, pp. 235–236] [3, pp. 240–241] [4, pp. 221–222].	Model the expected environment using tools such as SPENVIS [45] or CREME-MC [46] and select components whose test conditions exceed the predicted exposure. Athena uses a TMS570 device characterized to approximately 30 krad, while ORCASat selected a device tested to 6 krad [47, 8, 9].
Total ionizing dose (TID)	TID represents cumulative ionizing-radiation exposure and can gradually alter device parameters or cause permanent failure. NASA notes that many COTS parts may tolerate only approximately 2–10 krad, although tolerance varies strongly with device and fabrication process [1, pp. 177–178] [2, pp. 191–193] [3, pp. 194–196] [4, pp. 191–192].	Use components with tested TID ratings, such as the VORAGO VA41630 MCU or GR712RC processor, or apply shielding where COTS components cannot meet the required dose [48, 49].
Displacement damage dose (DDD)	Displacement damage results from non-ionizing particle interactions and must be considered separately from TID, particularly for sensors, optoelectronics, and some semiconductor technologies [1, p. 219] [2, p. 235] [3, pp. 240–241] [4, p. 221].	Require component-specific proton or neutron characterization rather than assuming that a TID rating also covers displacement damage. Microchip publishes separate proton, TID, and SEE characterization for its radiation-tolerant FPGA families [50].
Single-event upset (SEU)	An SEU changes a memory or logic state without necessarily causing permanent damage. Protection is required for processor state, working memory, non-volatile memory, and FPGA configuration [1, pp. 219–220] [2, pp. 235–236] [3, pp. 240–241] [4, pp. 221–222].	Use ECC or EDAC, memory scrubbing, redundant boot images, software consistency checks, and configuration-immune programmable logic. The GR712RC provides memory EDAC, while the RTG4 FPGA uses SEU-immune flash configuration [49, 51].
Single-event latch-up (SEL)	SEL may place a semiconductor device into a destructive high-current state. The affected circuit must be detected and disconnected before permanent damage occurs [1, pp. 219–220] [2, pp. 235–236] [3, pp. 240–241] [4, pp. 221–222].	Select SEL-immune components where practical and place vulnerable loads behind current-limited, resettable power switches. The 3D PLUS 3DPM0168 LCL provides overcurrent detection, load disconnection, and controlled repowering [48, 52].
COTS versus radiation-hardened components	COTS components generally provide higher performance, lower cost, and better availability, while radiation-hardened parts provide greater environmental assurance. Small spacecraft frequently use a hybrid approach [1, p. 219] [2, p. 235] [3, p. 240] [4, pp. 221–222].	Combine a COTS processor with radiation-tolerant supervisors, memories, programmable logic, or power-control circuitry. NEUDOSE uses a COTS AVR32 primary processor with a secondary Zynq-7000 system designed using both radiation-hardened and non-radiation-hardened components [10].
Watchdog and fault recovery	Radiation-induced faults may leave the processor or communication interface unresponsive. Recovery mechanisms should remain operational even when the primary processor has failed [1, p. 220] [2, p. 236] [3, p. 241] [4, pp. 219–220, 222].	Use an independent watchdog, reset supervisor, communication timeout, and individually switchable power domains. The GR712RC includes watchdog support, while external protection switches can independently cycle affected loads [49, 52].

Table 10: Protection considerations (Continued)

Consideration	Description	Example solutions
Memory protection	Processor memory, mass storage, boot images, and FPGA configuration can have different radiation responses and should be assessed independently. NASA identifies ECC, EDAC, scrubbing, and redundant storage as common mitigation methods [1, pp. 218–220] [2, pp. 234–236] [3, pp. 239–241] [4, pp. 221–224].	Use ECC-protected working memory, radiation-tolerant MRAM, redundant boot images, and TMR-protected non-volatile storage. Example devices include GR712RC EDAC-protected memory and radiation-tolerant MRAM and SPI NOR modules [49, 53, 54].
Redundancy and voting	Redundancy may be applied at the logic, processor, memory, board, or subsystem level. It improves fault tolerance but increases power, mass, synchronization, and verification requirements [1, p. 220] [2, p. 236] [3, p. 241] [4, pp. 219–222].	Use lockstep processors, cold-redundant OBCs, triple-modular redundancy, or distributed backup controllers. Athena and ORCASat use lockstep-capable safety MCUs, while 3D PLUS offers TMR-protected SPI NOR memory [8, 9, 54].
Radiation-hardened-by-design (RHBD)	RHBD improves radiation tolerance through hardened circuit cells, radiation-aware layout, specialized device structures, and selective hardening of vulnerable functions [4, p. 222].	Use RHBD devices such as VORAGO HARDSIL-based Cortex-M MCUs or Microchip RTG4 FPGAs with radiation-hardened registers and flash-based configuration [48, 51].
Structural and localized shielding	The spacecraft structure provides some inherent shielding, but protection depends on material, thickness, component placement, and shielding in every direction. Local shielding may be more mass-efficient than increasing the thickness of the entire structure [1, pp. 176–179] [2, pp. 190–194] [3, pp. 193–196] [4, pp. 190–193].	Place sensitive processors or memories within aluminium, titanium, graded-Z, or composite shields. Commercial localized shielding products, such as Plasteel, target high-performance COTS processors, cameras, and memories [55, 56].
Secondary-radiation effects	Increasing shield thickness does not always proportionally reduce dose. High-energy particles interacting with shielding may generate secondary particles, making material choice and layer ordering important [1, pp. 176–179] [3, pp. 193–196] [4, pp. 190–193].	Evaluate shielding using particle-transport simulation rather than aluminium thickness alone. Graded-Z shielding combines materials with different atomic numbers to reduce secondary radiation [56].
Surface charging and electrostatic discharge	Surface charging and electrostatic discharge can damage or disrupt electrical systems. This is separate from TID and single-event-effect protection [1, p. 179] [2, p. 194] [3, pp. 196–197] [4, p. 193].	Use conductive bonding, controlled grounding paths, charge-dissipative coatings, and appropriate PCB surface treatments. The Shields-1 experiment included a charge-dissipation film together with radiation-shielding technologies [56].

Conclusion

This report outlined the state-of-the-art OBC review from both internal design standpoint through the Design Space section as well as system integration standpoint through System Architecture section. Internal design is broken into selection of control unit, protocol set and interconnect topology, as well as peripheral array. Each of these components have a wide selection of components and designs which will need to be researched further and used as part of trades once requirements are established. Further research and collaboration with other subsystems is also required to guide the internal design decisions for this subsystem.

Appendix A: Academic OBCs

Name	Team	MCU	Protocols	Standard	Notes	Ref
STM32L4 PC/104 academic OBC	VST104 thesis design	STM32L4—ARM-based	CAN	PC104	pc 104 cubesat design.	[5]
Athena OBC	University of Alberta (Albertasat)	TI TMS570LC4357 (ARM Cortex R5F)			Embedded memory type: Flash. Features ideal for satellite OBC application, including radiation & space reliability (rad-hard, SEL immune 48 MeV at 125°C, LAT 30krad), high computational performance (lockstep dual-core compare, 300MHz CPU clock, 1.66DMIPS/MHz).	[6, 8]
Orcasat OBC	University of Victoria	TI TMS570LS0714 (Dual ARM Cortex-R4F)			Embedded memory type: Flash. Operates in lockstep, ECC integrated flash & RAM banks, radiation-tested up to 6 krad, 80MHz core clock.	[7, 9]
LORIS OBC	Dalhousie University	NXP i.MX6DL (Single-core ARM Cortex-A9)	USB 2.0; SPI; I2C; PWM; UART; GPIO; ADC		Embedded memory type: Flash. Low profile, power efficient SoC SBC. 4GB eMMC flash storage, 256MB DDR3 RAM, 800MHz operational frequency, -40°C to 85°C temperature range.	[11, 30]
NEUDOSE OBC	McMaster University	Primary: AVR32. Secondary: Xilinx Zynq-7000 SoC			Primary is COTS chosen for high TRL to reduce mission risk; secondary is designed for radiation tolerance using rad-hard and non-rad-hard components.	[10]
RADSAT-SK OBC	University of Saskatchewan	ARM926EJ-S			Embedded memory type: SDRAM. Uses ISISpace satellite-subsystems libraries, contains MMU, designed to interface with external SDRAM.	[25]

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